A Cloud-Based Data Leakage Prevention System Using Rule-Based Content Inspection

TEAM MEMBERS :

SABARIRAJA. S (620822205090)

RANJITH. P (620822205084)

SHANTHOSH. M (620822205098)

ROHITH. V (620822205088)

GUIDE : JAYANTHI. R,AP/IT

Problem Statement

- Users unknowingly upload sensitive information to cloud platforms
- Cloud providers cannot always detect confidential data correctly
- Existing DLP solutions rely on centralized inspection
- Centralized systems introduce privacy risks and single points of failure
- No simple mechanism exists to verify file integrity after upload
- A secure, lightweight, user-side DLP system is required

Objective

- To design a user-side Data Leakage Prevention system
- To detect sensitive data before cloud upload
- To prevent accidental and malicious data leakage
- To preserve user privacy during content inspection
- To verify file integrity using cryptographic hashing
- To maintain low computational overhead for real-time use

Existing System

- Data is uploaded directly to the cloud
- Security controls are applied mostly on the server side
- Cloud providers perform centralized content inspection
- Users have limited visibility and control
- Integrity checks are often absent or optional

Disadvantage

- Privacy loss due to server-side data inspection
- Dependency on cloud service providers
- Single point of failure in centralized systems
- Delayed detection of sensitive data leakage
- No reliable mechanism to detect file tampering

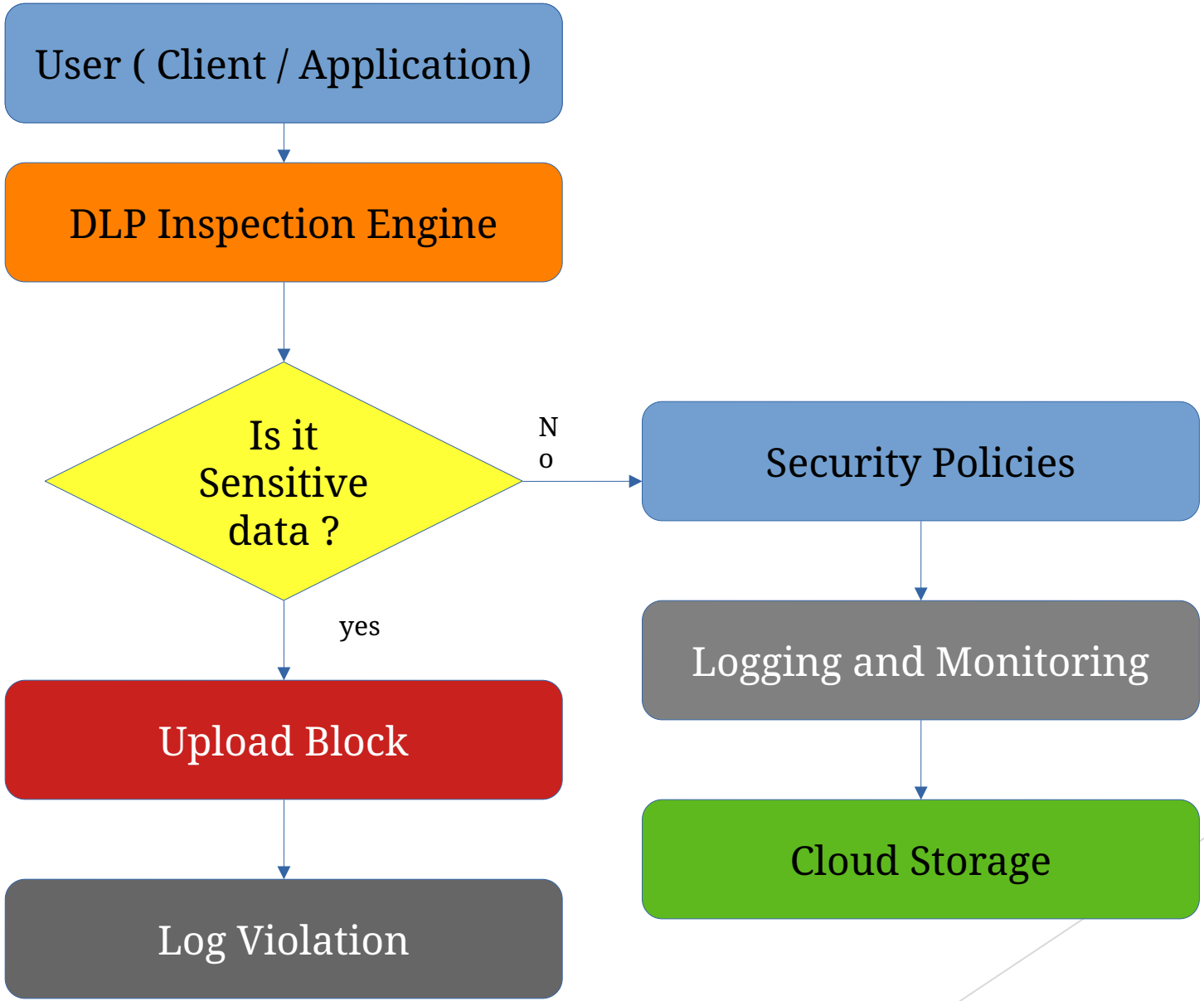
Proposed System :

- Data inspection performed on the user device
- Rule-based content scanning for sensitive patterns
- Upload blocked if sensitive data is detected
- Cryptographic hash generated for each file
- Hash used to verify file integrity and detect modifications

Advantage

- Prevents sensitive data leakage at the source
- Deterministic and transparent behavior
- Lightweight and suitable for real-time operation
- No dependence on third-party inspection
- Improves trust and security in cloud usage

SYSTEM ARCHITECTURE/FLOW DIAGRAM



Modules :

- Module 1: Rule-Based Content Inspection
- Module 2: Cryptographic Hashing for Integrity
- Module 3: Secure Violation Logging
- Module 4: User-Defined Security Policies
- Module 5: Performance Optimization & Cloud Integration

MODULE 1: Rule-Based Content Inspection

- Goal: Automatically detect sensitive data (PII, financial, credentials) in files and payloads
- Approach: Predefined rules using regex patterns, keywords, and context (e.g., “DOB: 01/01/2000”)
- Pipeline: Input → normalization → rule evaluation → list of violations → actions (block/mask/allow)
- Benefits: Simple, explainable, and auditable detection; foundation for all later modules

MODULE 2: Cryptographic Hashing for Integrity

- Goal: Make DLP records and content fingerprints tamper-evident
- Technique: Secure hash functions (e.g., SHA-256) applied to violation fields and/or content
- Design: Per-record hashes +
  optional hash chaining across log entries
- Outcome:
  Any modification of critical fields is detectable during audits or inve stigations

Module 3: Secure Violation Logging

- Goal: Capture DLP violations in a secure, compliant, and useful way
- Log contents: Timestamp, rule ID, user/service, resource info, action, severity, violation_hash
- Security: Integrity via hashes; confidentiality via masking/truncation and access control on logs
- Value: Reliable evidence for audits, incident response, and tuning DLP rules/policies

Conclusion

- The proposed blockchain-based voting system addresses major drawbacks of traditional electronic voting.
- Decentralization eliminates single points of failure and increases system reliability.
- Smart contracts enforce election rules automatically without human intervention.
- The system enhances transparency, security, and voter trust.
- The prototype demonstrates the practical feasibility of blockchain in electronic voting.
- This work lays the foundation for future large-scale and national-level implementations.

THANK YOU !